

Some number problems

1. Show that it is possible to arrange the numbers 1 to 15 in a line so that the sum of every pair of adjacent numbers is a perfect square.
2. Show that there is only one way to do this (other than reversing the list).
3. Show that it is not possible to do this for the numbers 1 to n when $n < 15$.
4. What is the smallest n for which there is more than one way (other than reversing the list) to arrange the numbers 1 to n so that the sum of every pair of adjacent numbers is a perfect square?